MINORITY DEGREES, MAJOR DOLLARS

How Minority Degree Attainment Shapes Public Institutional Revenues

Introduction

Recent action to repeal affirmative action (AA) and diversity, equity, and inclusion (DEI) within universities has significantly increased within the past few years. This can be attributed to various social and political movements, such as the US Supreme Court's overturning of affirmative action and Prop. 209 in Florida that abolished all public AA programs, strung together with the ideology that fairness, integrity, and academic validity will be restored if equal opportunity initiatives are dissolved.

Articles written by Brackes (2012), Howell (2010), and Hinrichs (2012) all find that the population of minority students fell with the bans on AA. To further that point, Howell (2010) reports that underrepresented minority student enrollment within the most selective universities in the US is expected to decrease by 10 percent. Hinrichs (2012) observes that minority enrollment shifts from more selective to less selective institutions after AA bans. So, there is a distinguishable difference between university student populations post-AA bans.

My objective is to conduct an in-depth analysis of the efficacy of DEI initiatives at public universities, with a particular focus on how increased enrollment and degree attainment among Black and Hispanic students influence institutional outcomes.

Research Question and Hypotheses

Research Question: What is the effect of minority degree attainment on public university revenue from 2002 to 2022?

H1: Higher degree attainment of Black and Hispanic university students will predict greater university revenues.

H0: Black and Hispanic degree attainment does not predict a significant change in revenue.

(Additional Analysis: Examine descriptive statistics for HBCU and PWI revenue and interpret any significant differences.)

Methodology

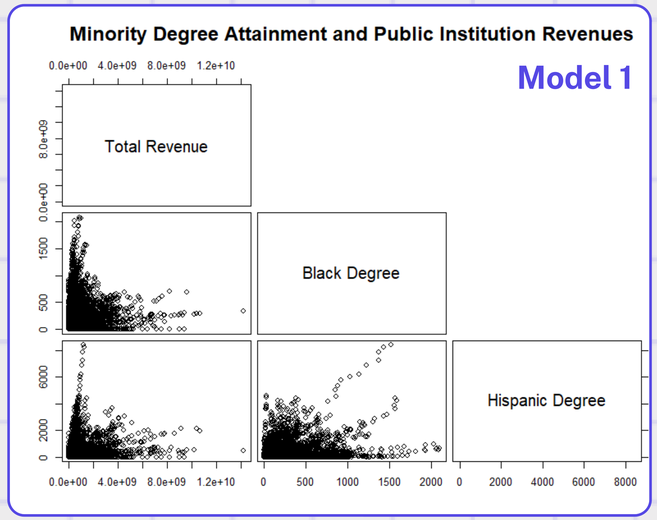
To conduct the analysis, I collected IPEDS data on Black and Hispanic enrollment percentages and institutional revenue for each year from 2002 to 2022. Then, to conduct my analysis I employed multiple linear regression provides coefficients (slope, y-intercept, etc.) that quantify the relationship between each predictor variable and the outcome. This allowed for a clearer understanding of how much each variable contributes to revenue changes, offering more precise insights than correlation alone. Multiple regression will allow me to incorporate additional variables (e.g., enrollment size, state funding, and tuition) and assess how these factors influence the model's explanatory power.

There are two models used to conduct these analyses. Model 1 contains only Black and Hispanic degree attainment numbers as the predictor variables and total institutional revenue as the outcome. Model 2 expanded this framework by including the following additional predictors: total enrollment, federal funding, student aid percentage, and in-state tuition. Which allows us to see how these additional variables will contribute to the explanatory power that these analyses provide.

HBCUs vs. PWIs

An additional analysis of revenue per student for public Historically Black Colleges and Universities (HBCU) and Predominantly White Institutions (PWI) in 2022 revealed significant disparities. Public HBCUs generated an average of $41,785 per student, with a relatively symmetric revenue distribution (Median = $42,262) and moderate variability (SD = $12,461). This indicates consistent funding patterns among HBCUs, though differences persist between institutions.

PWIs, on the other hand, averaged $209,245 in revenue per student, with a much lower median of $28,695. The significant gap between the mean and median highlights a right-skewed distribution, where a small number of institutions with extremely high revenues pull the average upward. This skewness reflects the influence of large, well-funded universities with extensive research budgets, substantial endowments, or high tuition fees. The high variability in revenue per student underscores the unequal distribution of resources among PWIs, contrasting with the relatively uniform pattern observed in HBCUs. Additionally, there are substantially fewer HBCUs than PWIs, which could contribute to the significant differences in spread.

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Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      194476726    5937561   32.75   <2e-16 ***
blkhisrev$Black_Degree     524295      28133   18.64   <2e-16 ***
blkhisrev$hispanicdegrees  459132      13336   34.43   <2e-16 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 667800000 on 16818 degrees of freedom
Multiple R-squared:  0.1228,    Adjusted R-squared:  0.1227
F-statistic:  1177 on 2 and 16818 DF,  p-value: < 2.2e-16
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Formula and Calculation of Multiple Linear Regression

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_p x_{ip} + \epsilon$$

where, for $i = n$ observations:

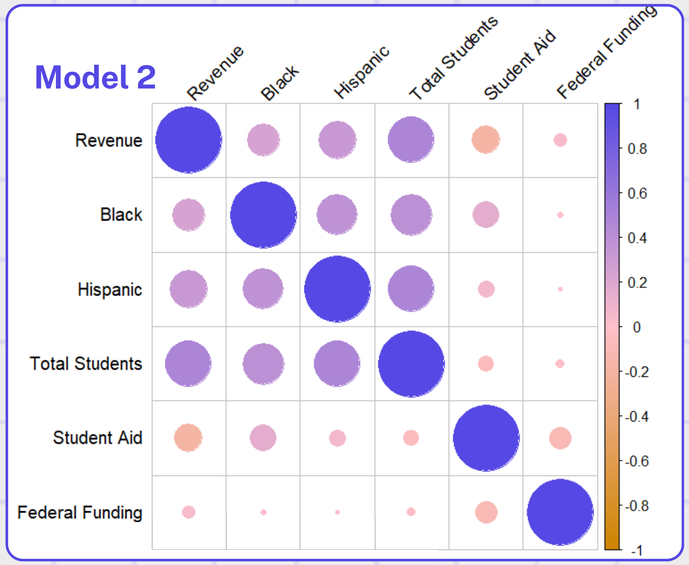
y_i = dependent variable

x_i = explanatory variables

β_0 = y-intercept

β_p = slope coefficients for each explanatory variable

ϵ = the model's error term (also known as the residuals)




```
Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)    -1.027e+08  1.631e+07  -6.296 3.16e-10 ***
fedfund         1.250e+00  1.694e-01   7.380 1.67e-13 ***
totstudent      4.154e+04  5.913e+02  70.260  < 2e-16 ***
studentaidpercent -5.645e+06  2.830e+05 -19.947  < 2e-16 ***
instate         4.704e+04  1.633e+03  28.812  < 2e-16 ***
blkdegree       1.967e+06  3.022e+05   6.509 7.85e-11 ***
hisdegree       3.123e+05  4.423e+05   0.706     0.48
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 604100000 on 13274 degrees of freedom
  (3540 observations deleted due to missingness)
Multiple R-squared:  0.3654,    Adjusted R-squared:  0.3652
F-statistic:  1274 on 6 and 13274 DF,  p-value: < 2.2e-16
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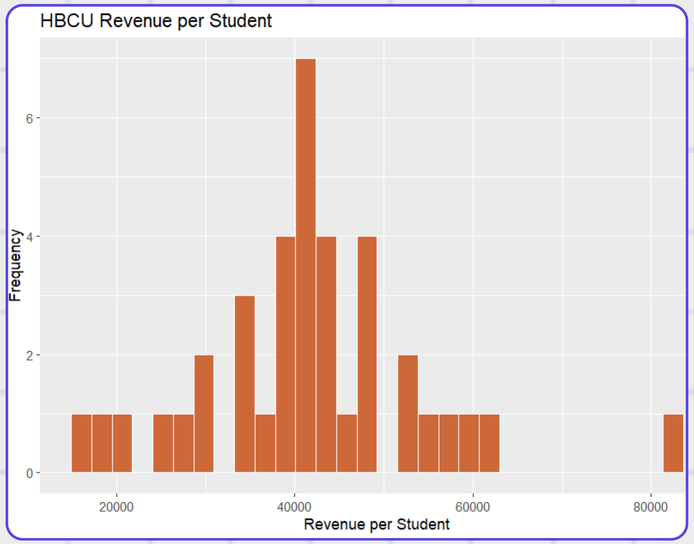

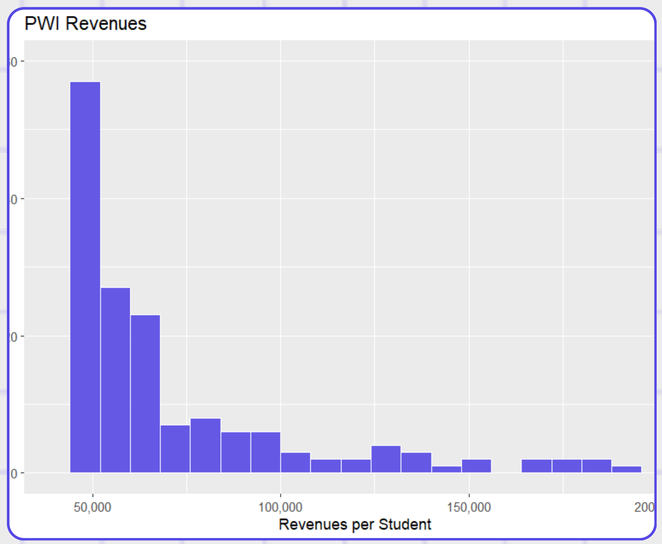

Degree Attainment and Revenue

In Model 1, the intercept (β_0 = $19,447,672) indicates that even in the absence of Black and Hispanic degree attainment, public institutions are estimated to generate approximately $19.4 million in revenue. Both predictors—Black degree attainment (β_1 = $524,295) and Hispanic degree attainment (β_2 = $459,132)—were statistically significant, showing that for every additional degree attained by Black and Hispanic students, institutional revenue increases by $524,295 and $459,132, respectively. This model explains 12.3% of the variance in revenue (R^2 = 0.1228), suggesting a modest fit. However, the residual standard error ($667.8 million) highlights limitations in predicting institutional revenue, reflecting the complexity of revenue generation. The model's overall significance ($p < 2.2e-16$) underscores the importance of these variables.

In Model 2, the intercept (β_0 = -$102,708,000) suggests that in the absence of all predictors, institutional revenue would theoretically fall to a deficit—though this value is less meaningful due to the inclusion of multiple variables. Black degree attainment (β_5 = $1,966,690) remained a strong, statistically significant predictor, indicating that each additional degree attained by Black students contributes nearly $2 million to revenue. Conversely, Hispanic degree attainment (β_6 = $312,305; p = 0.48) became statistically insignificant. Among the added predictors, total student enrollment (β_2 = $41,541) and in-state tuition (β_4 = $47,039) emerged as major contributors to revenue, while student aid percentage (β_3 = -$5,644,893) showed a significant negative relationship. This expanded model explained 36.5% of the variance (R^2 = 0.3654) and showed improved accuracy with a lower residual standard error ($604.1 million).

Finally, stepwise regression using AIC refinement confirmed the significance of key predictors from Model 2. Black degree attainment remained influential (β_5 = $1,966,691), while Hispanic degree attainment had minimal explanatory power. Predictors such as total enrollment and in-state tuition continued to dominate. This final model achieved the lowest AIC (574,761) and demonstrated reliability, with metrics like RMSE ($603.9 million) and MAE ($296.8 million) reinforcing its predictive accuracy.

Conclusion

Several factors may explain the moderate explanatory power of the models. Potential confounding factors include state and local policies, economic conditions, and private donations or endowments, which likely play significant roles. Moreover, another confounding variable is larger universities (inherently making more money due to size) attracting more Black and Hispanic students. Further study may be required and encouraged to substantiate these possible connections.

An interesting finding was the considerable negative impact of student aid percentage on revenue, suggesting a need to investigate the interaction between financial aid and institutional funding. Again, future research could explore the racial demographics of students receiving financial aid to understand these dynamics better. This may uncover whether systemic disparities in financial aid distribution contribute to broader revenue trends in higher education.

Sources

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